CHAPTER 7:

Bounce

> HAMLET: O God, I could be bounded in a nutshell, and count myself a king of infinite space, were it not that I have bad dreams.
>
> William Shakespeare, *Hamlet*

On September 14, 2015, at 9:50 a.m. and 45 seconds UTC, you were, for the briefest moment, just a little bit taller.

The gravitational wave crest that washed through you had been traveling across the cosmos, warping space itself in its wake, for 1.3 billion years, ever since it was set off by the violent merging of two black holes each 30 times more massive than the Sun. You might not have noticed the boost—after all, you grew by less than one millionth the width of a proton—but physicists at the Laser Interferometry Gravitational-Wave Observatory (LIGO) did. The first detection of gravitational waves was the culmination of a decades-long search, requiring the development of new technologies and the creation of the most sensitive equipment in the history of experimental physics. Finally detecting those ripples in spacetime was heralded as the ultimate vindication of Einstein's general theory of relativity.

But even more significantly, it was the dawning of a new age of astronomical observation. It opened up the universe to a totally new way of seeing. Instead of collecting light or high-energy particles from distant sources, we could now reach out and feel the vibration of space itself, creating for the first time a window onto the kind of distant cosmic violence that can shake the very foundations of reality.

Since that first discovery, gravitational wave astronomy has continued to show us the inspirals and catastrophic mergers of black holes and neutron stars and allowed us to study the workings of gravity with an unprecedented level of precision. But gravitational waves may hold the key to something even more fundamental. They may give us a new view of the shape and origin of our universe, and present us with the possibility of determining